GATESYNC

Smart Residential Society Security & Real-Time Visitor Management System

A full-stack, installable Progressive Web Application (PWA) designed for modern gated residential societies. It replaces manual paper logbooks with instant mobile push notifications for residents, 1-tap courier clearance, pre-approved QR guest passes, and society maintenance accounting.

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PROJECT TYPE & ACADEMIC YEAR

Final Year Capstone Engineering Project

Academic Year 2025–2026

TECHNOLOGY FOCUS

WebSockets (STOMP), Spring Boot, React 19 PWA, MongoDB Atlas

Introduction & Motivation

Why Gated Communities Need a Modern Digital Gate Security System

TRADITIONAL PAPER REGISTER PROBLEMS

- **Long Gate Queues at Peak Hours**
  Guards manually write down visitor names, phone numbers, and flat numbers in a physical book. Each car takes 2–4 minutes, causing entry traffic.
- **Zero Privacy for Residents & Visitors**
  The paper logbook sits open at the security cabin. Anyone standing at the gate can read or photograph resident phone numbers and visiting flats.
- **Unanswered Intercoms & Calls**
  Guards frequently call residents on intercom or phone to verify visitors. When calls are missed, guards let visitors in unverified or keep them waiting outside.
- **High Volume of E-Commerce Deliveries**
  Daily delivery couriers (Swiggy, Zomato, Amazon, Blinkit) create huge manual overhead for guards who have to log repetitive entries.

THE GATESYNC DIGITAL SOLUTION

- **Instant Mobile Push Alerts (< 150ms)**
  When a visitor arrives, the guard enters details and an instant pop-up alert appears on the resident phone with visitor info and photo.
- **1-Tap Resident Remote Approvals**
  Residents simply tap Approve or Deny from their smartphone anywhere (at home, office, or travelling) without phone calls.
- **Pre-Approved Guest Passes with QR Code**
  Residents create guest passes beforehand for friends or family. Guests show their QR code or 6-digit OTP for 2-second fast clearance.
- **1-Tap Fast Delivery Partner Entry**
  One-click buttons for Swiggy, Zomato, Amazon, and Blinkit allow guards to clear couriers without typing names manually.

System Architecture & Technologies Used

Full-Stack Web & Mobile Architecture Built for Reliability, Speed and Security

FRONTEND CLIENT TIER (Progressive Web App)

React 19, Vite, Lucide Icons, STOMP over SockJS, Service Worker

Responsive, mobile-friendly interface installable on Android and iOS. Includes client-side offline queueing to store entries if the gate internet is down.

BACKEND APPLICATION TIER (REST & WebSocket Server)

Java 21 LTS, Spring Boot 3.3.4, Spring Security 6, JJWT

Stateless microservice handling user authentication, role-based access control, WebSocket message broker (/topic and /user/queue), and transactional business logic.

DATABASE & PERSISTENCE TIER

MongoDB Atlas (Cloud NoSQL) & H2 In-Memory Relational Engine

MongoDB Atlas persistently stores visitor logs, device push tokens, guest passes, and audit trails. In-memory relational mapping is used for fast user lookup.

CLOUD HOSTING & DEPLOYMENT

Render Cloud (Dockerized Spring Boot) & GitHub Pages (PWA Hosting)

Containerized production backend deployed on Render with automated health check (/api/health) and static frontend deployed on GitHub Pages.

Society Admin Dashboard & Live Overview

Centralized Web Portal for Society Office Bearers and Committee Members

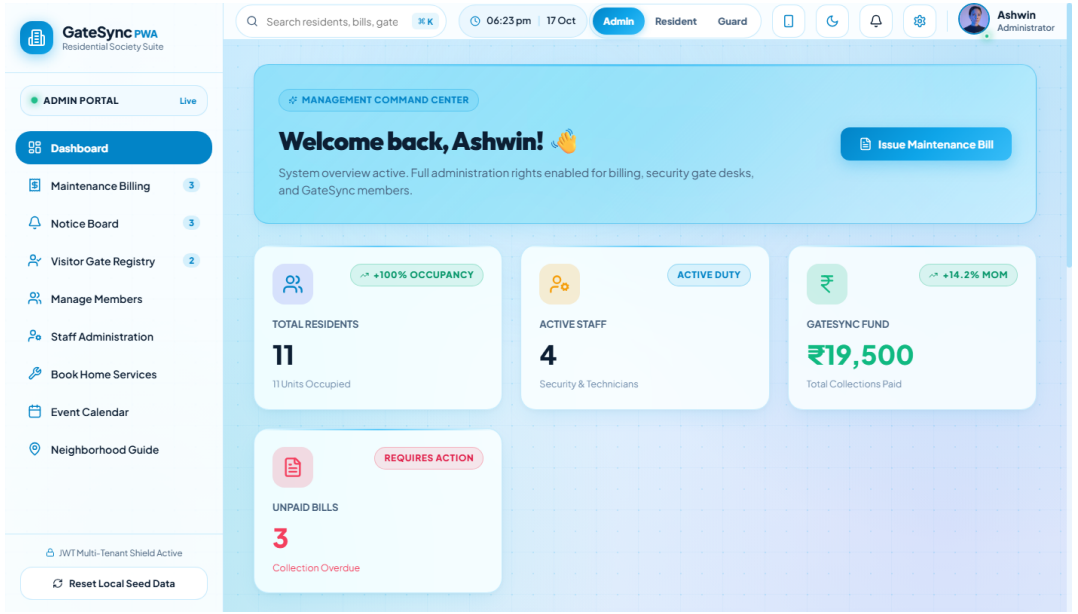
KEY FEATURES & USER WORKFLOW

- **Real-Time Society Statistics**
  Displays overall counts of flats (128), residents (342), active security guards, and total visitor traffic for the day.

- **Live Gate Traffic Feed**
  Shows visitors currently inside the society, pending gate approvals, and recent checkouts with instant live updates.

- **Quick Access Actions**
  One-click shortcuts to add/edit residents, manage security personnel, issue circulars, and review maintenance fees.

- **Secure Role Separation**
  Admins have full oversight but sensitive resident passwords are encrypted and individual privacy is strictly preserved.



Guard Terminal: Fast Visitor Registration

Simple, Touch-Friendly Interface Designed for Security Guards on Tablets and Phones

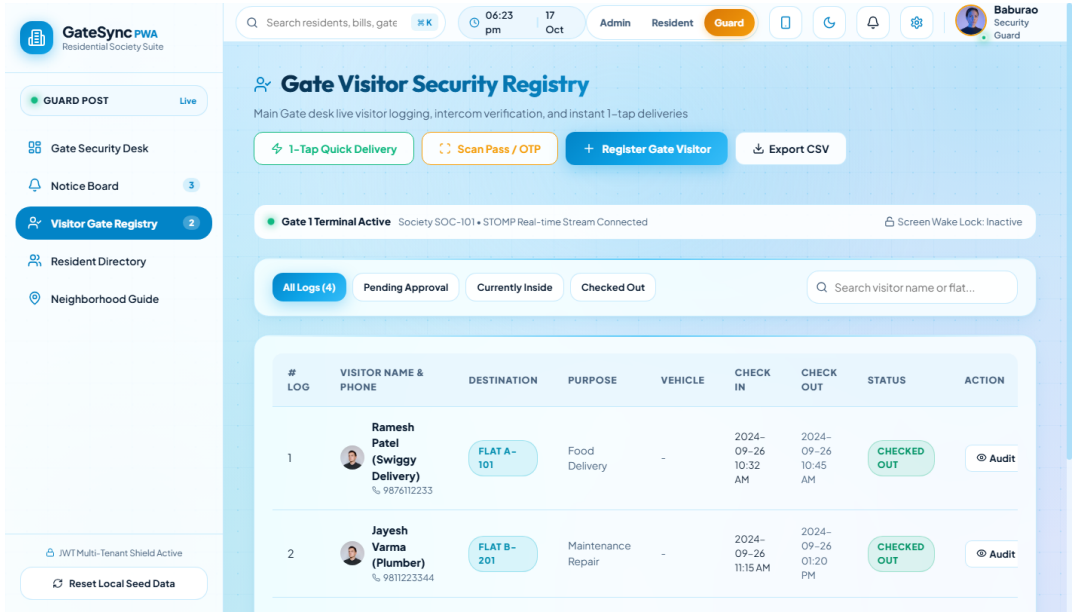
KEY FEATURES & USER WORKFLOW

- **Quick 15-Second Registration**
  Guard enters the visitor full name, mobile number, flat number, visit purpose, and vehicle registration number.

- **Fast Auto-Complete Flat Search**
  Guard types flat number (e.g., A-101) and the resident details appear instantly without looking up phone numbers.

- **Instant Push to Resident Mobile**
  Clicking Send Entry Request pushes the visitor notification directly to the resident phone over WebSockets.

- **On-Screen Status Tracking**
  The guard screen shows status badges: Waiting for Approval, Approved (Green), or Denied (Red) in real-time.



Resident Real-Time Alert & 1-Tap Approval

Instant Pop-Up Alert on Resident Smartphone for Fast Remote Entry Decisions

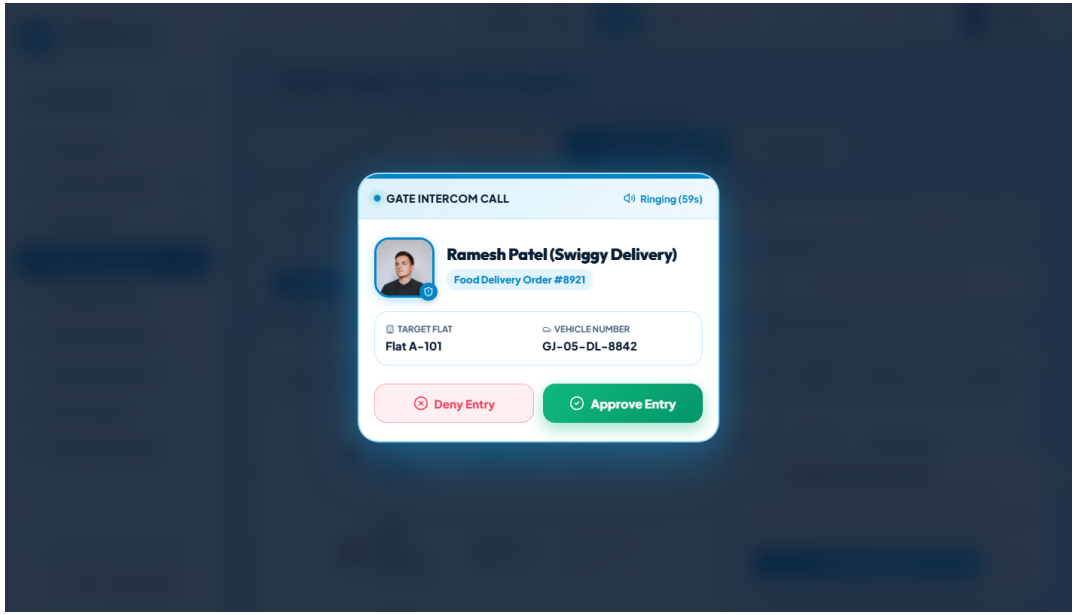
KEY FEATURES & USER WORKFLOW

- **Automatic Screen Pop-Up Modal**
  When the guard logs a visitor, a clean modal immediately pops up on the resident screen accompanied by an alert chime.

- **Complete Visitor Information**
  Shows the visitor photo, full name, visit reason, vehicle number, and gate location for clear identification.

- **1-Tap Binary Decision**
  Resident taps the Green Approve Entry button or Red Deny Entry button. A reason can be added if denying entry.

- **Prevents Conflicting Decisions**
  If two residents in the same flat view the alert, the first response is recorded and prevents double-approvals.



1-Tap Fast Delivery & Courier Clearance

Fast-Track Entry for Popular Logistics and Food Delivery Couriers

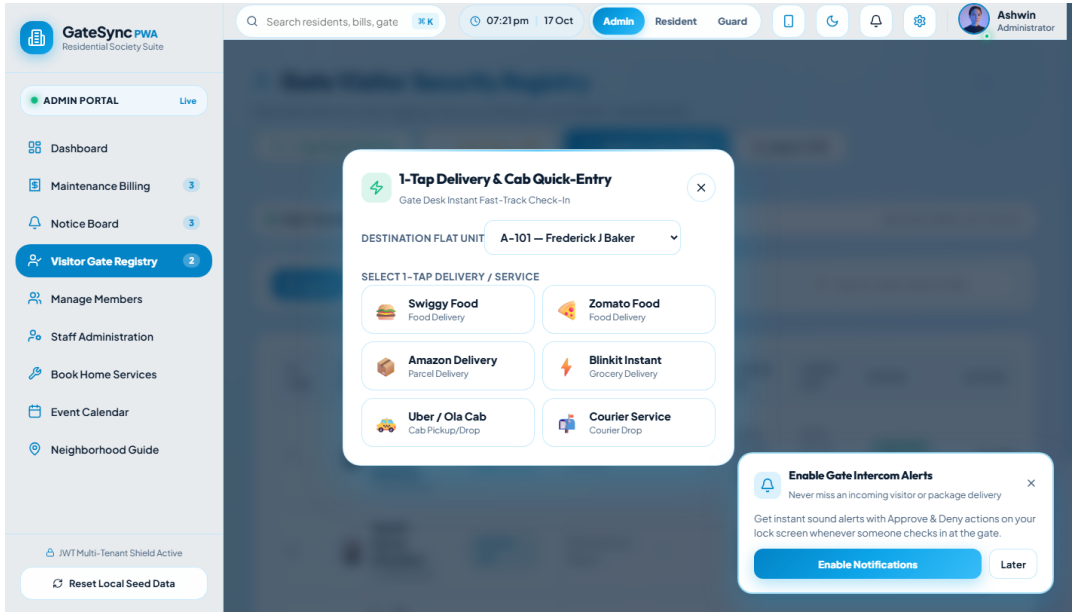
KEY FEATURES & USER WORKFLOW

- **One-Tap Delivery Brand Buttons**
  Dedicated buttons for Swiggy, Zomato, Amazon, Blinkit, and Flipkart allow guards to clear couriers without typing.

- **Reduces Gate Bottlenecks**
  Cuts courier entry processing from 2 minutes down to under 5 seconds during busy lunch and dinner hours.

- **Doorstep vs Gate Delivery Options**
  Residents can instruct whether the delivery person is permitted to come upstairs or leave the parcel at the gate.

- **Searchable Delivery Records**
  Every courier entry is logged with provider name, target flat, and timestamp for parcel tracking.



Pre-Approved Digital Passes & QR Code Scanner

Hassle-Free Entry for Family, Friends, Parties, and Expected Service Workers

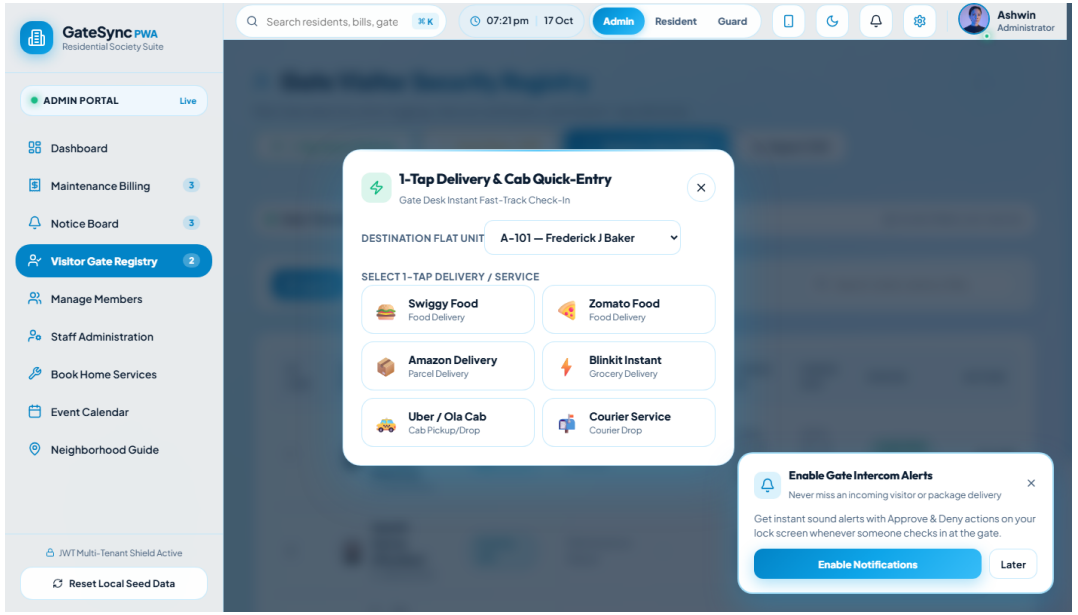
KEY FEATURES & USER WORKFLOW

- **Residents Create Passes in Advance**
  Residents enter guest details and pick valid dates/times (e.g., valid today from 5:00 PM to 10:00 PM).

- **Generates 6-Digit Code & QR Code**
  A secure 6-digit numeric pass code and scannable QR code are automatically generated for the guest.

- **1-Click WhatsApp Sharing**
  Resident can directly share the invite message with the QR code and entry link to their guest via WhatsApp.

- **Camera QR Scan at Gatehouse**
  Guard scans the guest QR code with the tablet camera or enters the 6-digit code for instant 2-second entry.



Digital Visitor Registry & Historical Gate Logs

Searchable Digital Ledger of Every Person Entering and Leaving the Campus

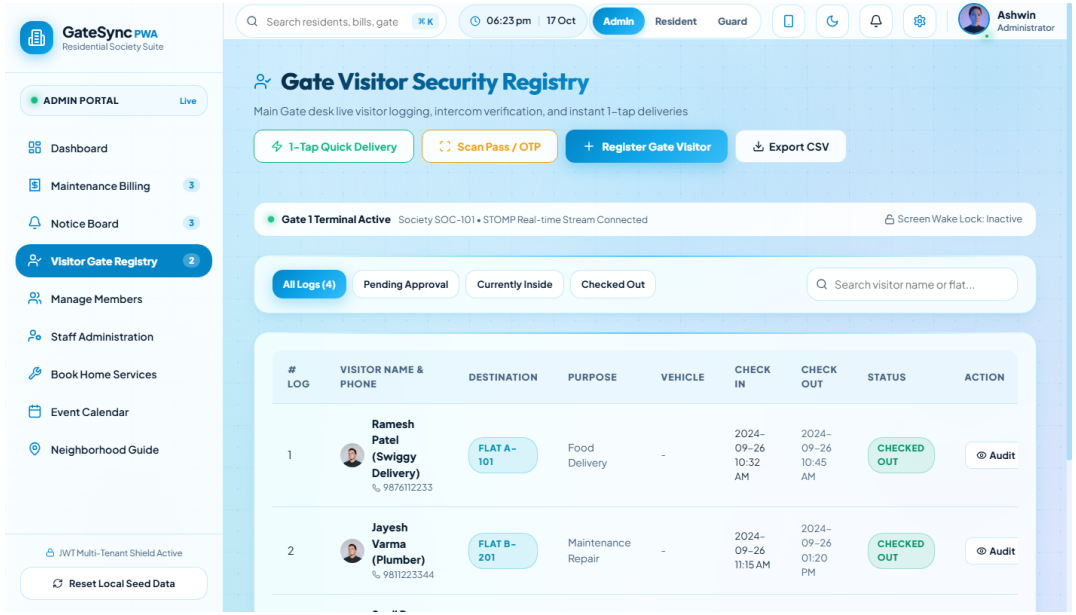
KEY FEATURES & USER WORKFLOW

- **Complete Real-Time Gate Registry**
  Searchable table showing visitor names, mobile numbers, vehicle numbers, entry time, and visiting flat.

- **Clear Visual Status Badges**
  Indicates who is currently INSIDE the society, who has CHECKED OUT, and which requests were REJECTED.

- **1-Tap Departure Checkout**
  When a visitor leaves the campus, guards click Check Out with one click, recording exact exit time and duration of stay.

- **Easy Search & Security Export**
  Admins can search past logs by date range, vehicle number, or flat for security reviews or police inquiries.



09. SOCIETY FINANCE

Society Maintenance Billing & Problem Helpdesk

Automated Monthly Billing, Dues Tracking, and Resident Repair Requests

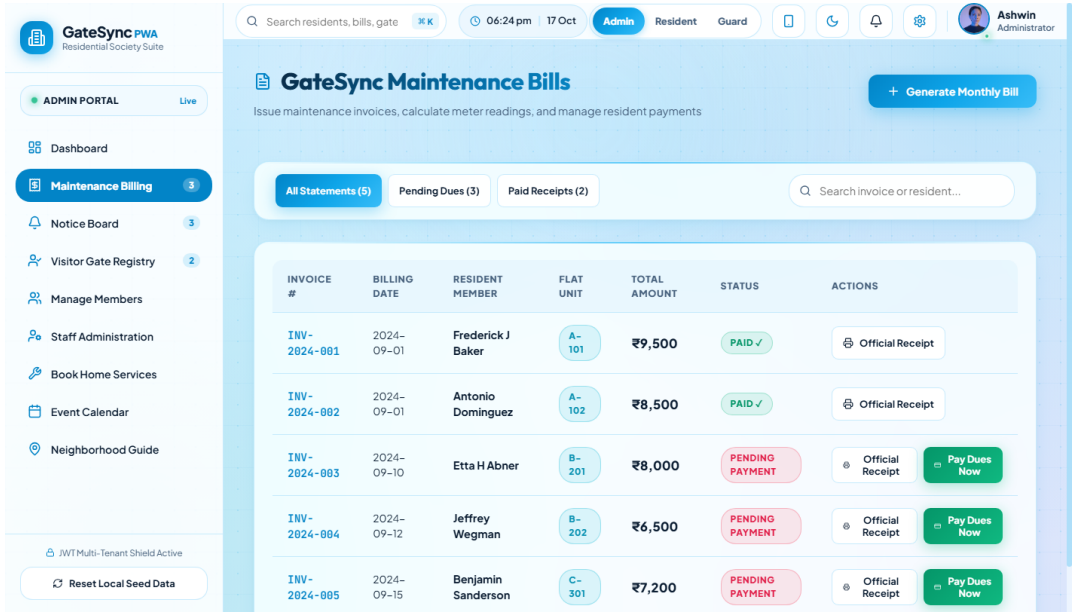
KEY FEATURES & USER WORKFLOW

- **Flat-Wise Maintenance Invoicing**
  Generates monthly society dues with clear breakdown for maintenance, parking, water, and sinking funds.

- **Payment Status Tracking**
  Tracks which flats have Paid, are Pending, or Overdue with simple color-coded badges for society managers.

- **Printable Digital PDF Invoices**
  Residents and committee members can view and download itemized PDF payment receipts.

- **Resident Maintenance Helpdesk**
  Residents can raise society maintenance tickets (e.g., plumbing leaks, elevator issues) with live progress tracking.



Security Architecture & Offline Reliability

How GateSync Protects Resident Data and Keeps Working When Internet Drops

USER AUTH & DATA PRIVACY

- Zero-Trust JWT Authentication: 15-minute rotating session tokens keep user accounts secure.
- Encrypted Passwords: Passwords hashed with BCrypt; default passwords force a reset on first login.
- Strict Role-Based Access: Admin, Resident, and Guard roles only access authorized screens.
- Privacy by Design: Visitor phone numbers and personal resident data are hidden from public view.

OFFLINE-FIRST GATE CONTINUITY

- Service Worker Caching: The web app starts up and opens even if the gate Wi-Fi is disconnected.
- Offline Entry Queue: Visitor entries recorded without internet are stored safely in local browser memory.
- Automatic Reconnection Sync: Once internet connectivity restores, queued entries sync automatically.
- No Lost Visitor Logs: Guards never have to fall back to paper registers during network blips.

CLOUD DATABASE & BACKUPS

- MongoDB Atlas Cloud Cluster: Multi-tenant database hosted in the cloud with automated replication.
- 2-Second Alert Deduplication: Prevents double-notification spam if a button is accidentally clicked twice.
- Emergency SOS Siren: Residents or guards can trigger campus panic alert with immediate guard siren.
- Tamper-Proof Audit Trail: Every admin and security action logs timestamp and actor identity.

THANK YOU

Questions & Project Evaluation Discussion

PROJECT LIVE DELIVERABLES & CLOUD ENDPOINTS

- **Live Web Application (PWA):** https://arpitthakor28.github.io/sms1/ (Installable responsive frontend)
- **Live Cloud Backend API (Render):** https://sms1-c9sj.onrender.com (Spring Boot 3 + Java 21 container)
- **Continuous Health Check (Nap Stopper):** https://sms1-c9sj.onrender.com/api/health (Uptime monitoring endpoint)
- **GitHub Source Code Repository:** https://github.com/arpitthakor28/sms1 (Complete code, documentation & poster)

STUDENT DEVELOPER

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